Harmonic Analysis Project Feedback

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This is brief feedback for the project on the sharp A_p bound of the Hardy-Littlewood maximal operator by Simon Gustafsson.

I found the project to overall be well structured and clear. The theory is presented in an understandable and motivated way and the proofs are clear and easy to follow. The main criticism I have is that the main point of the project, which is to present the proof in Lerner's paper, may to some extent be lost by the extent to which weights and weighted maximal functions is covered. It would be good if more was done to elevate Lerner's proof so that a reader could more easily focus on that. Other than this I only found some minor complaints. In the proof of Lemma 7 the following is written:

$$\left(\frac{1}{Q(x,r)}\int_{Q(x,r)}|f|dz\right)^{p-1} =$$

$$=\frac{w(Q(x,r))\sigma(Q(x,3r))^{p-1}}{|Q(x,r)|^p}\cdot\frac{1}{w(Q(x,r))}\int_{Q(x,r)}\left(\frac{1}{\sigma(Q(x,3r))}\int_{Q(x,r)}|f|dz\right)^{p-1}dy.$$

I did not find it clear enough were the outer integral comes from. Something else to note is in the definition of a dual weight. It is not mentioned that, because of the way things are defined, a dual weight is not necessarily a weight and this could cause confusing for the reader. For example in the definition itself the notation $\sigma(E)$, where σ is the dual weight of some weight and E a measurable set, is defined, which would be redundant unless σ is not necessarily a weight.